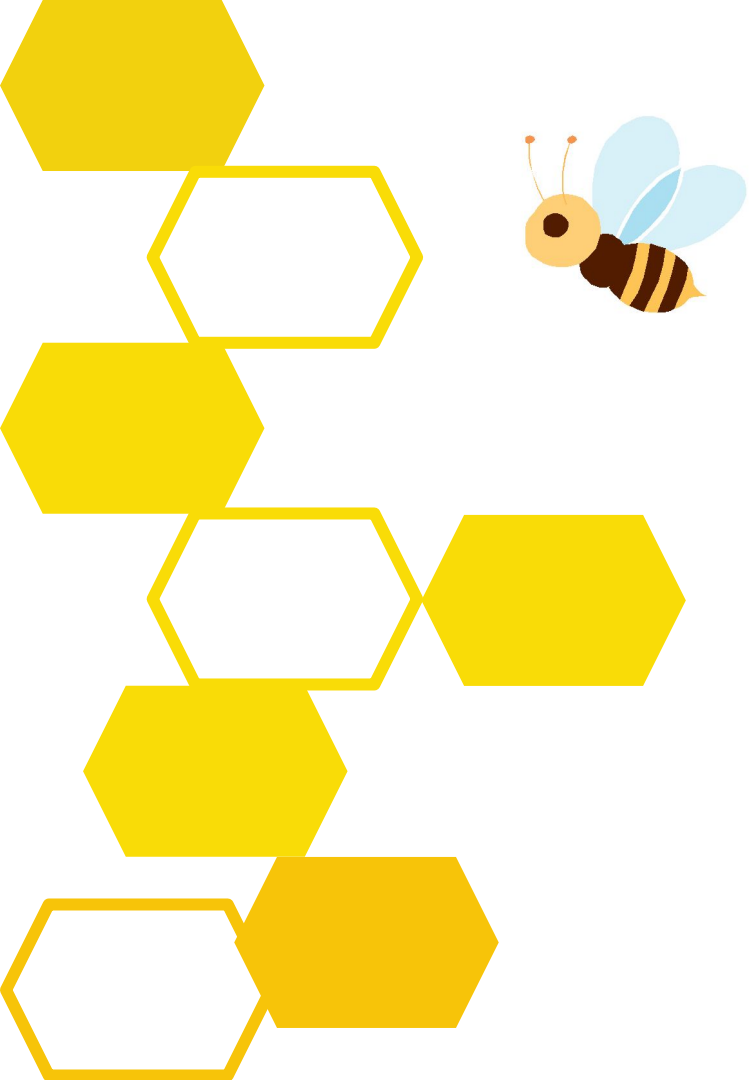




CDS 101

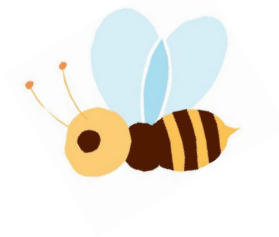
Final Project

Group 3

Junseo Choi, Sojeong Lee, Soyeon Park,
Ho-Jae Song, Heewon Yang



Table of contents



1. Introduction

2. Dataset

3. EDA

4. Hypothesis Testing

5. Modeling

6. Conclusion

7. Implication





Introduction

Background Information

These days increasing carbon emissions are giving serious impact on honeybees.

Honeybees help maintain tree species and reproduce among plants. In addition, honeybees play an essential role for the reproduction of plants as one of the main means of pollinating.

Carbon emissions also cause tornado more frequently in the United States. Our group focused on the correlation between carbon emissions, tornado frequency, and honey bees population.

As a result, we would like to find external factors for honey bees population decline.





Introduction

Why it is interesting

Research Question :

"Does the increase in honey bee loss percentage in the United States from 2015 to 2021 show a causal relationship with the increase in carbon emissions and a higher frequency of tornado occurrences?"





Dataset

Background Information

1. **Save the bees :** Data from honey bees in the United States from 2015 to 2022. These data were collected by the USDA and curated by me into a manageable dataset for EDA and predictions.
2. **CO2 Emissions USA :** Detailed insights into carbon dioxide emissions across various sectors and states in the United States, designed for data analysts interested in climate trends and sustainability.
3. **US Tornado Dataset 1950–2021 :** This dataset was derived from a dataset produced by NOAA's Storm Prediction Center.



Dataset

Source

1. Save the bees

< Selected Columns >

state: state within the USA.

quarter: Q1 is Jan to March, Q2 is April to June, Q3 is July to September, and Q4 is October to December

year: year between 2015 and 2022.

max_colonies: max number of honey bee colonies for that quarter.

percent_lost: percentage of honey bee colonies lost during that quarter.



Dataset

Source

2. CO2 Emissions USA

< Selected Columns >

year: The year of the emissions data.

State-name : The name of the state where the emissions were recorded.

Fuel-name : The type of fuel responsible for the emissions (e.g., Coal, Petroleum, Natural Gas).

Value : The amount of carbon dioxide emissions (presumably in million metric tons).



Dataset

Source

3. US Tornado Dataset 1950–2021

< Selected Columns >

yr : 4–digit year

st : State where tornado originated; 2–digit abbreviation





Dataset

Data Wrangling

1. Loading dataset
2. Processing by importing, tidying data to relabel and reorganize data
3. Transforming data by manipulating it to make it suitable for analysis.





Dataset

Data Wrangling



1. Load three datasets.
2. Filter year 2015–2021 to use same period.
3. Select columns.



Dataset

Data Wrangling

<Honey bee>

1. Select useful data to calculate average max colonies with quarter.
: bee_avg_max
2. Pivot wider into q1 ~ q4.
remaned (q1 = 1, q2 = 2, q3 = 3, q4 = 4)
3. Mutate average max colonies.





Dataset

Data Wrangling

<CO2 emission>

1. Sum of each fuel name values.
: emission_fuel
2. Pivot wide each fuel name.





Dataset

Data Wrangling



<Tornado>

1. Count tornado frequency by year and state.
: tornato_freq
2. Fill 0 at no tornado year and state.



Dataset

Data Wrangling

<Merge Data for Modeling>

Step 1: Bees vs Emission

1. Data Cleaning
bee state / emission state
2. Data Merge
bee_emission_df



Dataset

Data Wrangling

<Merge Data for Modeling>

Step 2: Bees vs Tornado

1. Data Cleaning

tornado_state

2. Data Merge

bee_state, tornado_state -> bee_tornado_df





Dataset

Data Wrangling

<Merge Data for Modeling>

Step 3: Bees vs Emission + Tornado

1. Data Cleaning

`bee_tornado_df / bee_emission_df`

2. Data Merge

`bee_df`

1. **Summary Statistics:** Provided mean, standard deviation, minimum, and maximum values for each dataset.

Maximum Colonies

- mean
- median
- minimum
- maximum
- SD (standard deviation)

```
print(avg_max_col_summary)
```

```
## avg_max_col_mean avg_max_col_sd avg_max_col_min avg_max_col_max
## 1      51.33017      46.94803      5.275      221.75
## avg_max_col_median
## 1      30.104
```

```
print(all_fuel_summary)
```

```
## all_fuels_mean all_fuels_sd all_fuels_min all_fuels_max all_fuels_median
## 1      168.1837      105.2404      28      422      139.5
```

Carbon Emission

- mean
- median
- minimum
- maximum
- SD (standard deviation)



1. **Summary Statistics:** Provided mean, standard deviation, minimum, and maximum values for each dataset.

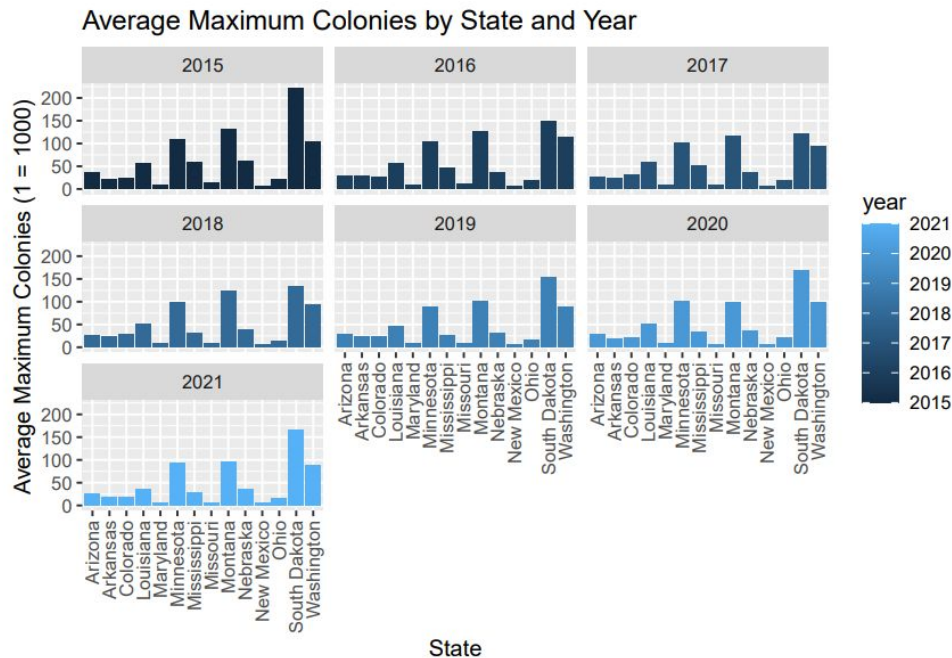
Tornado Count

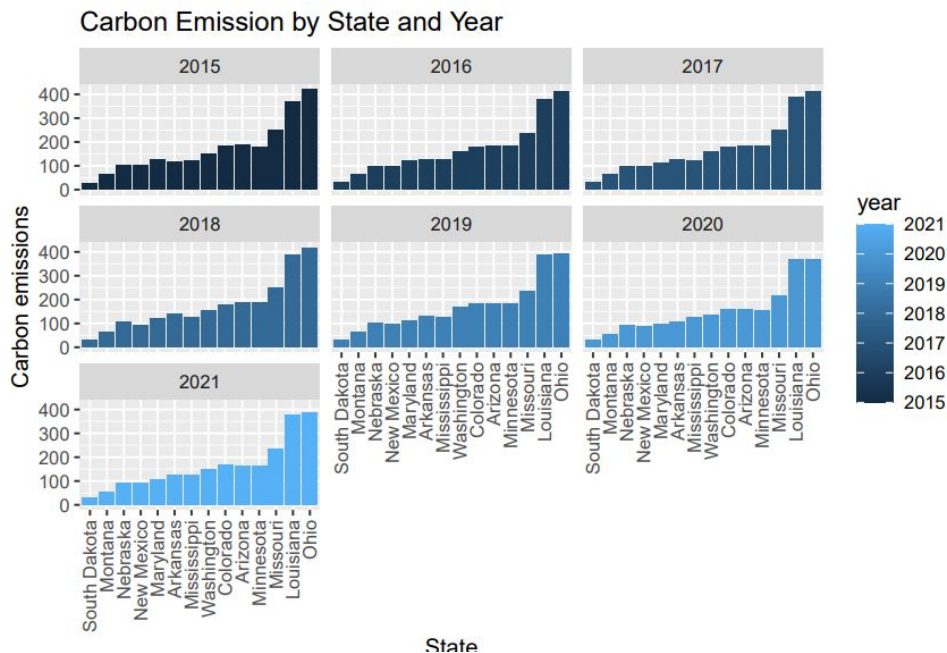
- mean
- median
- minimum
- maximum
- SD
(standard deviation)

```
print(tornado_summary)
```

```
## tornado_count_mean tornado_count_sd tornado_count_min tornado_count_max
## 1          28.80612          24.31946          1          113
## tornado_count_median
## 1          23
```





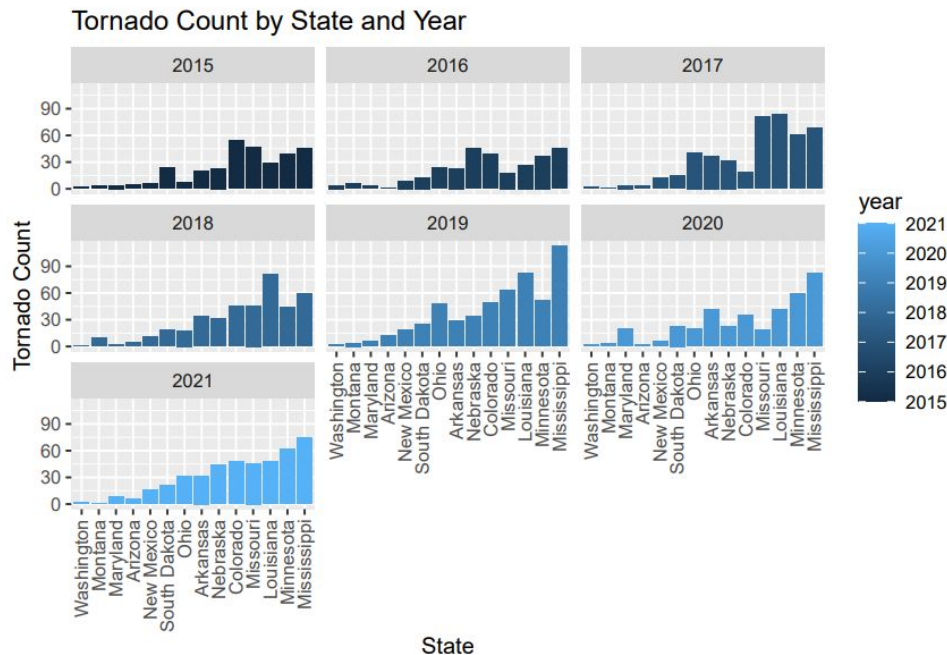


Yearly trend

- 2015 – 2016
- 2017
- 2018 – 2019
- 2020
- 2021

Ohio, Louisiana





Yearly trend

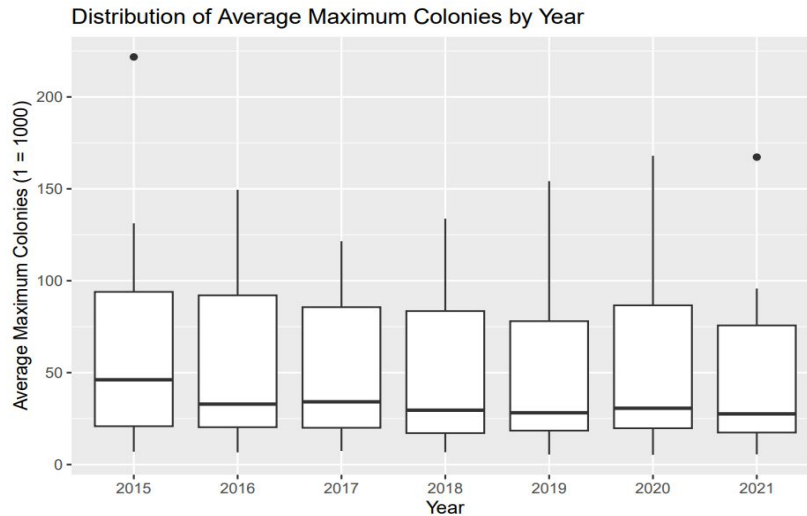
- 2015 – 2016
- 2017
- 2018 – 2019
- 2020
- 2021

Mississippi, Louisiana



EDA

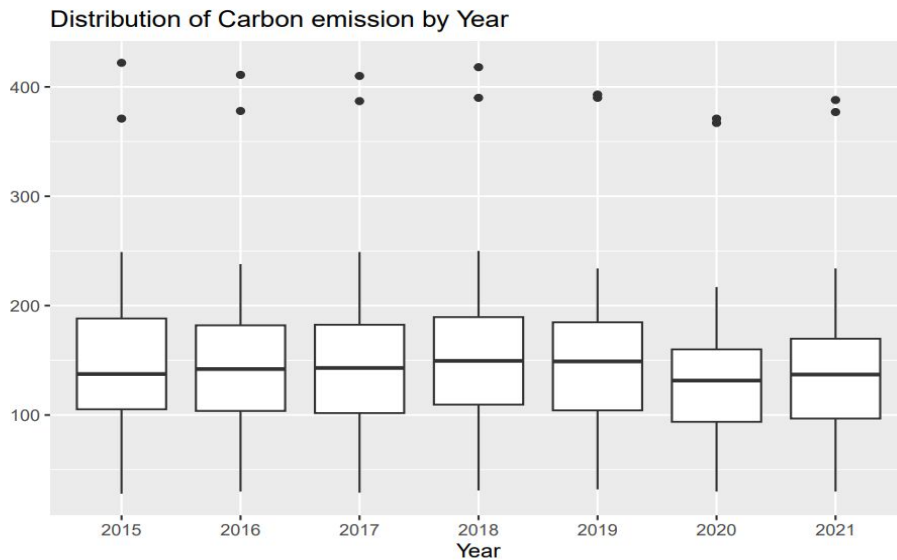
Visualization



- Median
- Skewness
- Outliers

EDA

Visualization

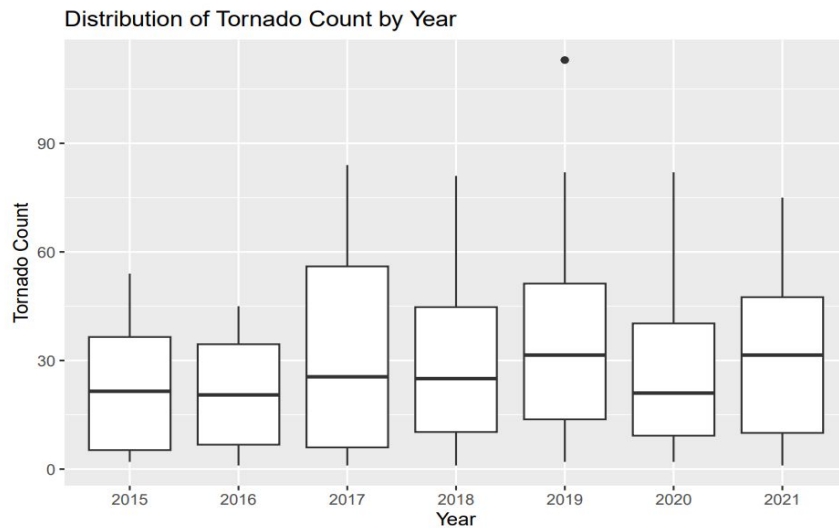


- Median
- Skewness
- Outliers



EDA

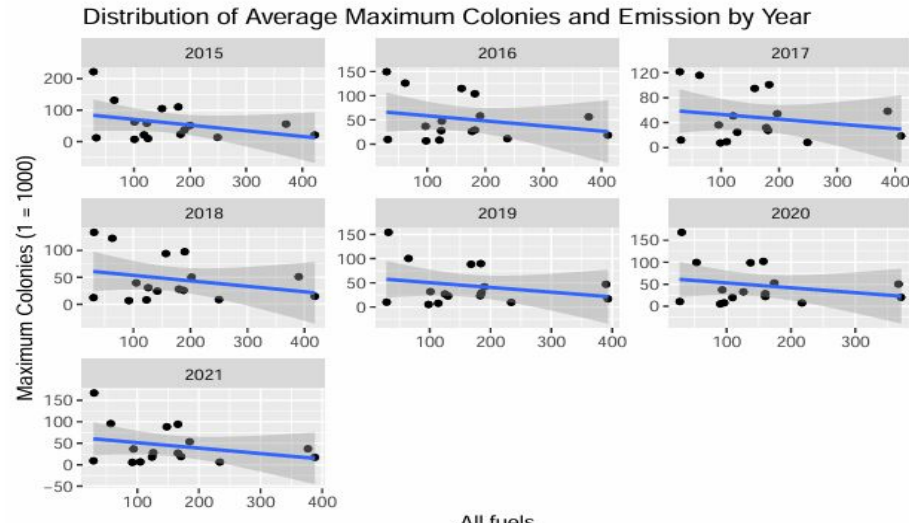
Visualization



- Median
- Skewness
- Outliers

EDA

Visualization

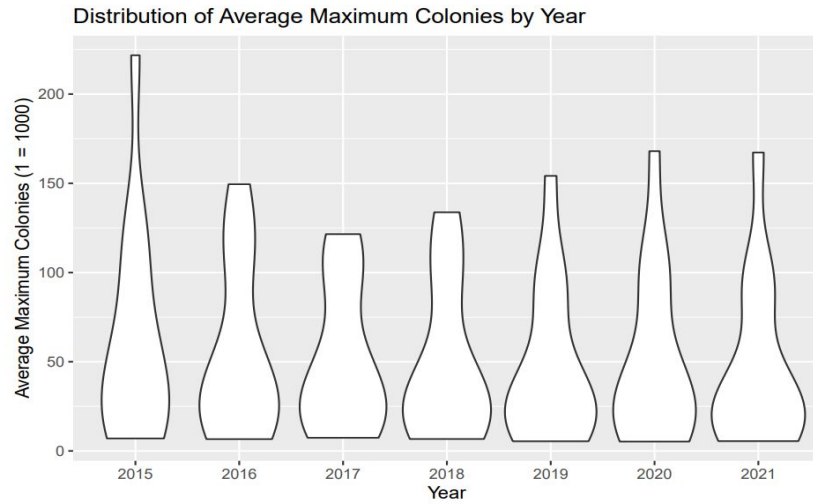


- Confidence interval
- Outliers
- Plot



EDA

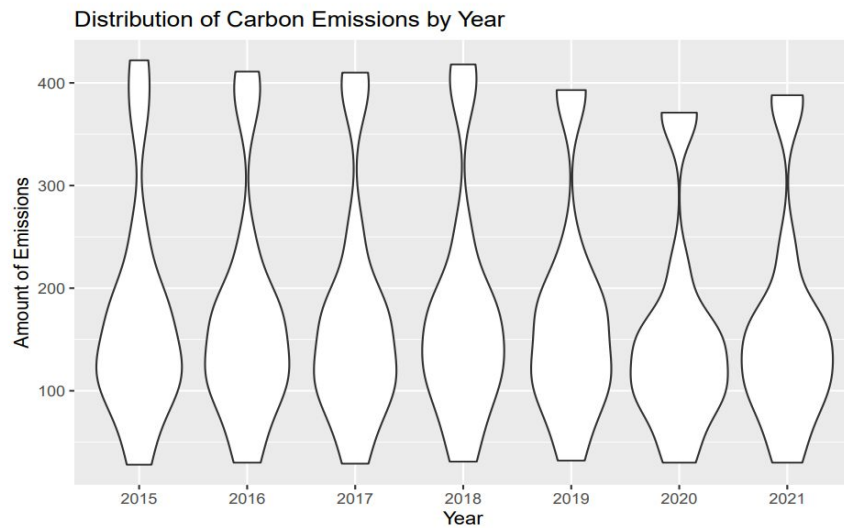
Visualization



- Median
- Density
- Outliers

EDA

Visualization

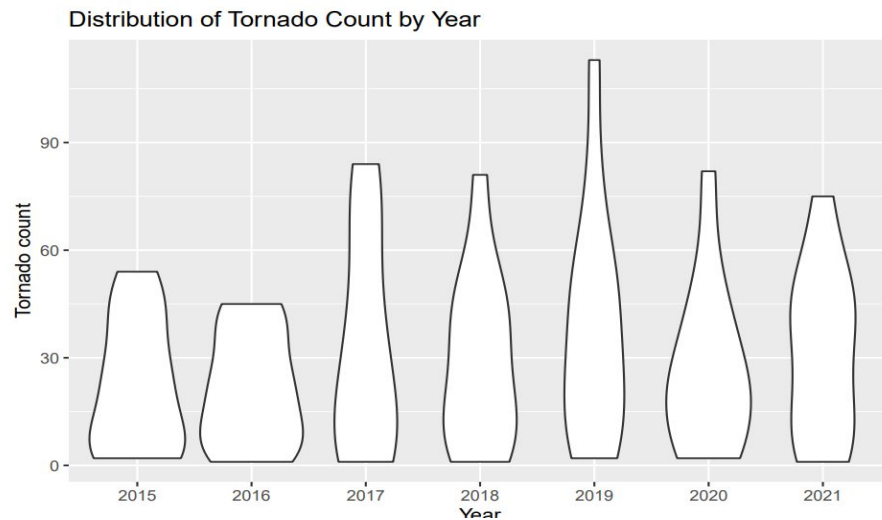


- Median
- Density
- Outliers



EDA

Visualization



- Median
- Density
- Outliers



Hypothesis Testing

presentation of hypothesis

Presentation of hypothesis :

- **Null Hypothesis (H_0):** There is no significant causation between carbon emissions and tornado occurrences with the decline in bee populations.
- **Alternative Hypothesis (H_1):** There is a significant causation between carbon emissions and tornado occurrences with the decline in bee populations.

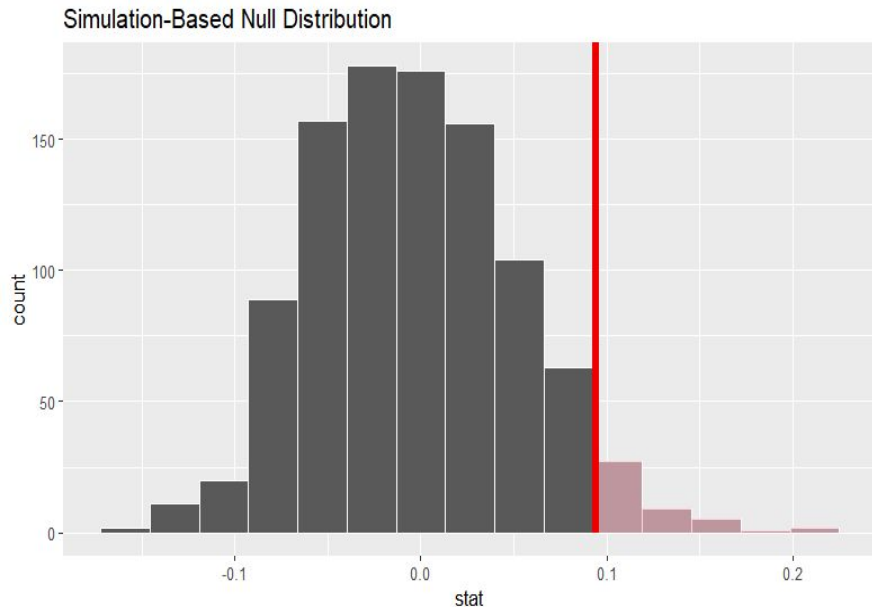


Hypothesis Testing

Interpretation of results

Graph Explanation

- **Simulation-Based Null Distribution:** This graph represents the null distribution generated through simulation.
- **Red Line (Observed Statistic):** The red line represents the observed statistic and shows where it falls within the null distribution.
- **Shaded Area (P-value Region):** The red shaded area represents the p-value region, indicating the proportion of simulated statistics that are more extreme than the observed statistic.



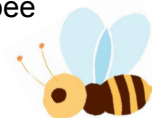
p_value
<dbl>

0.043

P-value Interpretation

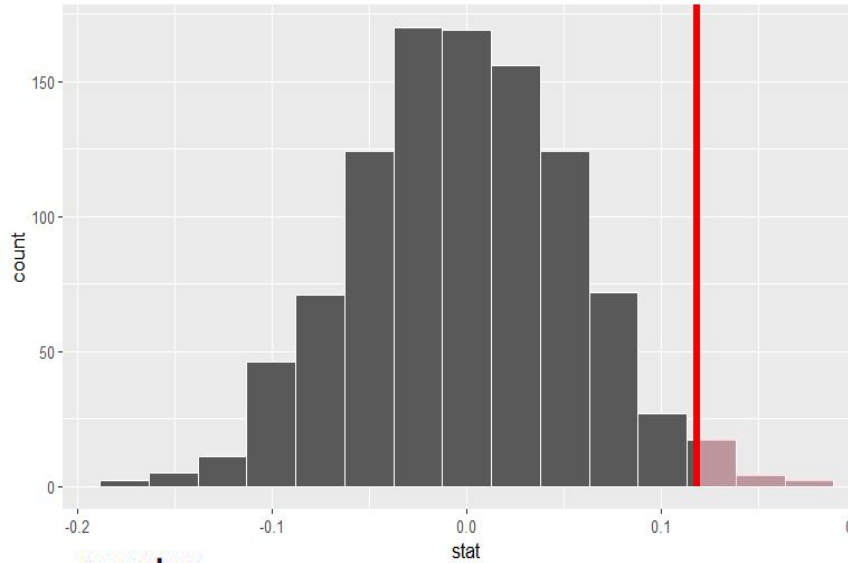
- A p-value of 0.043 is less than the commonly used significance level (α) of 0.05. Therefore, we can reject the null hypothesis (H_0) and accept the alternative hypothesis (H_1).
- This indicates that there is a significant causation between carbon emissions and the decline in bee populations.

Conclusion: There is a significant causation between carbon emissions and the decline in bee populations.



Hypothesis Testing

Simulation-Based Null Distribution



p_value
<dbl>

0.015

P-value Interpretation

- A p-value of 0.015 is less than the commonly used significance level (alpha) of 0.05. Therefore, we can reject the null hypothesis (H_0) and accept the alternative hypothesis (H_1).
- This indicates that there is a significant causation between tornado occurrences and the decline in bee populations.

Graph Explanation ## Interpretation of results

- **Simulation-Based Null Distribution:** This graph represents the null distribution generated through simulation.
- **Red Line (Observed Statistic):** The red line represents the observed statistic and shows where it falls within the null distribution.
- **Shaded Area (P-value Region):** The red shaded area represents the p-value region, indicating the proportion of simulated statistics that are more extreme than the observed statistic.



Conclusion: There is a significant causation between tornado occurrences and the decline in bee populations.

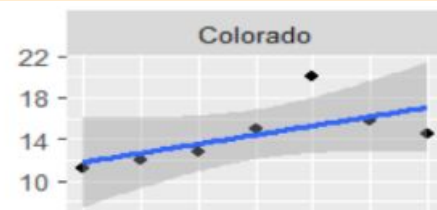
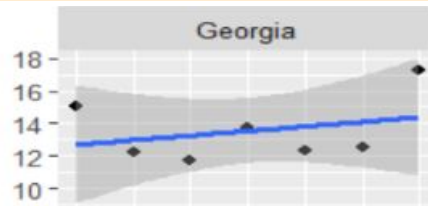
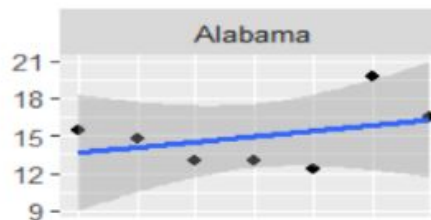
Modeling

Table of result

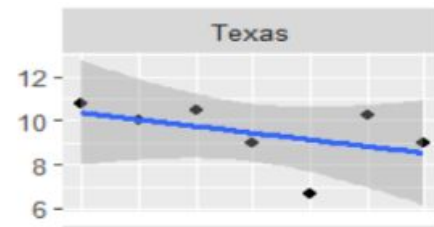
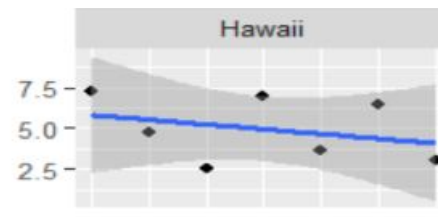
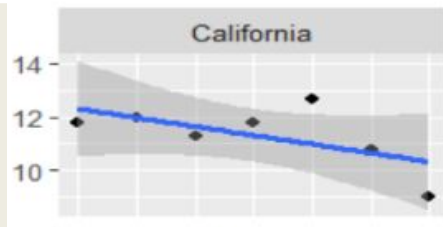
Step 1

- Filter states where percent lost of colonies were increased during 2015~2021
- Alabama, Arizona, Colorado, Georgia, Idaho, Maine, Maryland, Mississippi, Missouri, Montana, Virginia

**Increased
2015~2021**

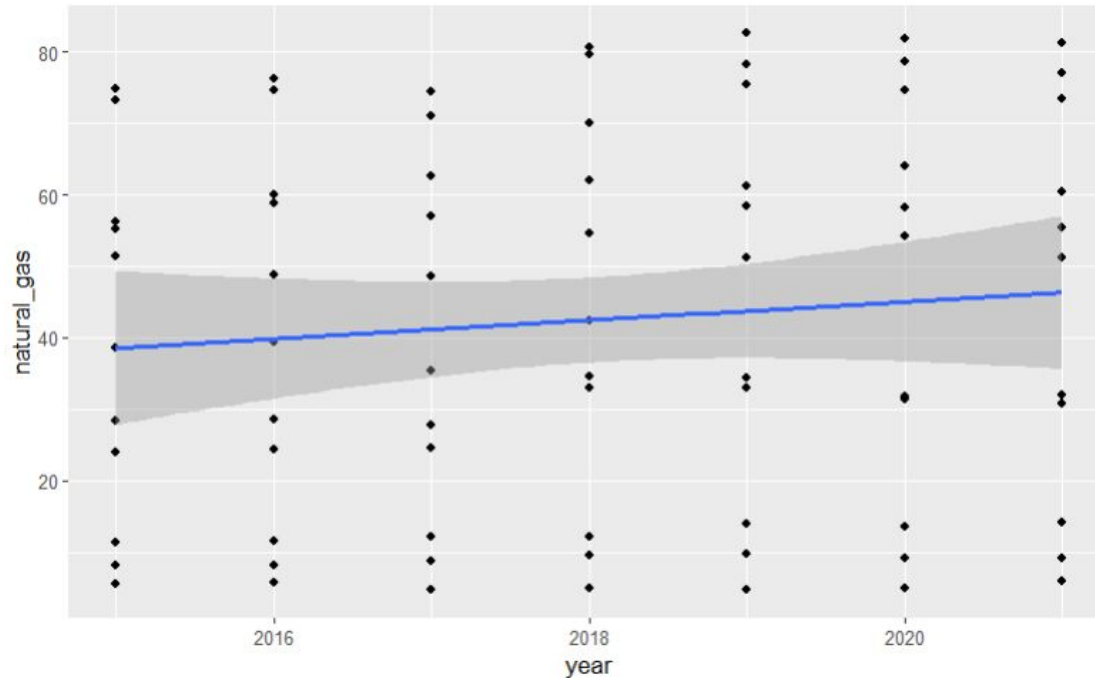


**Decreased
2015~2021**



Step 1

- ➔ When we look at the data of each type of fuel, the variable 'Natural Gas' has shown an increase in emissions from 2015 to 2021.





Step 2

[Build Model 1]

- bee_model_1
- Linear Regression Model
- Responsible Variable: **Average lost colonies percentage**
- Explanatory Variable: **Natural Gas**

```
# modeling 1  
bee_model_1 <- lm(avg_lost_col ~ natural_gas , data = new_bee_df_1)
```

[Linearity Condition]

- tidy() -> Y-intercept and Slope
- glance() -> R2 (r. squared)

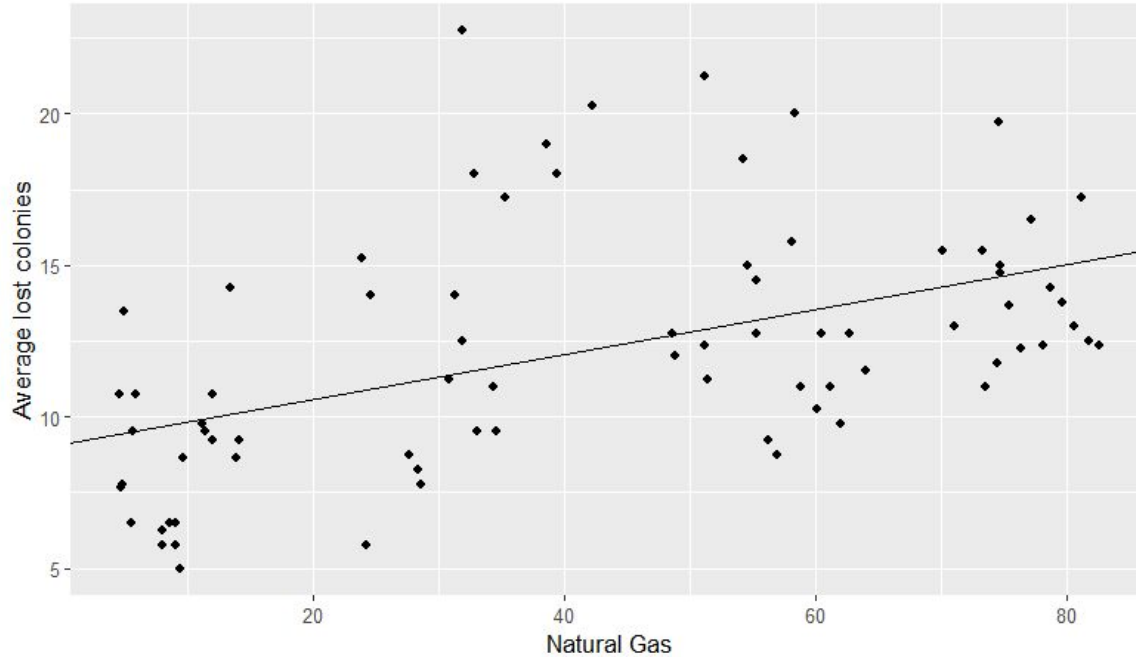
r.squared <dbl>
0.2351593

[Residuals Distribution]

- add_residuals()

Step 3

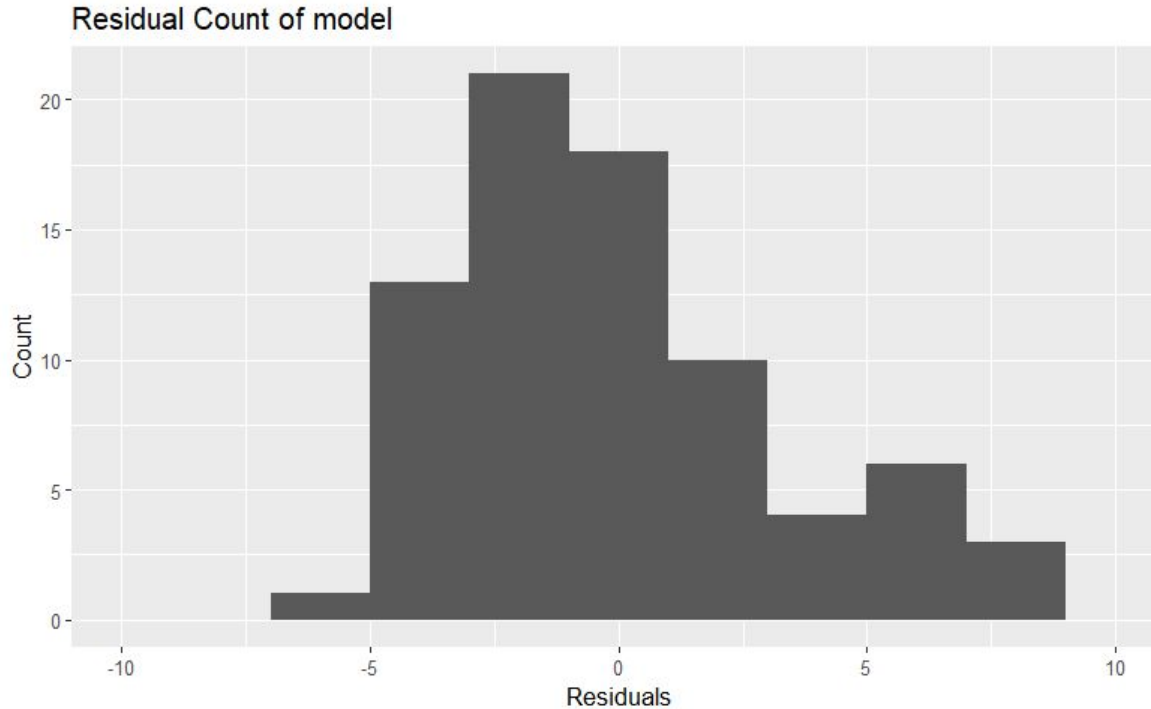
Linearity condition of model 1



- Positive linear trend.
- Natural Gas emissions increase
- Percentage of bee colony loss also increases.
- Points considerably spread.



Step 3



-Normal distribution

-Right-skewed

-Residuals concentrated between
[-5, 5]

-Have outliers on right side



Step 1

[Multiple linear regression model]

- y1: **Average lost colonies percentage**
- x1: **Natural Gas**
- x2: **Count**
- Two way interaction -> $x_1 * x_2$

$$\hat{y} = b_0 + b_1X_1 + b_2X_2 + b_3X_1X_2$$



Step 2

[Build Model 2]

- bee_model_2
- Multiple Linear Regression Model
- Responsible Variable **y: Average lost colonies percentage**
- Explanatory Variable **x1: Natural Gas, x2: Count**

```
# modeling 2
bee_model_2 <- lm(avg_lost_col ~ natural_gas + count + natural_gas * count,
                  data = bee_df_2)
```

[Linearity Condition]

- tidy() -> Y-intercept and Slope
- glance() -> R2 (r. squared)

r.squared

<dbl>

0.2983744

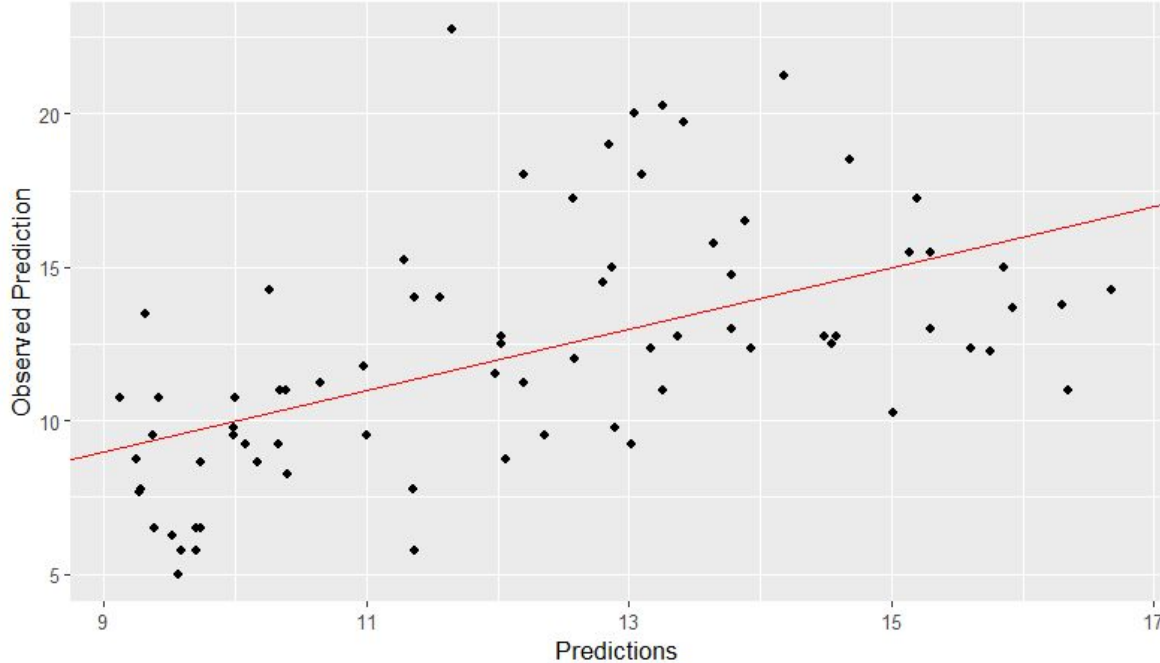
[Distribution]

- add_prediction()
- add_residuals()



Step 3

Observed Prediction of factors of model 2



-Observed values tend to increase as the predicted values increase.

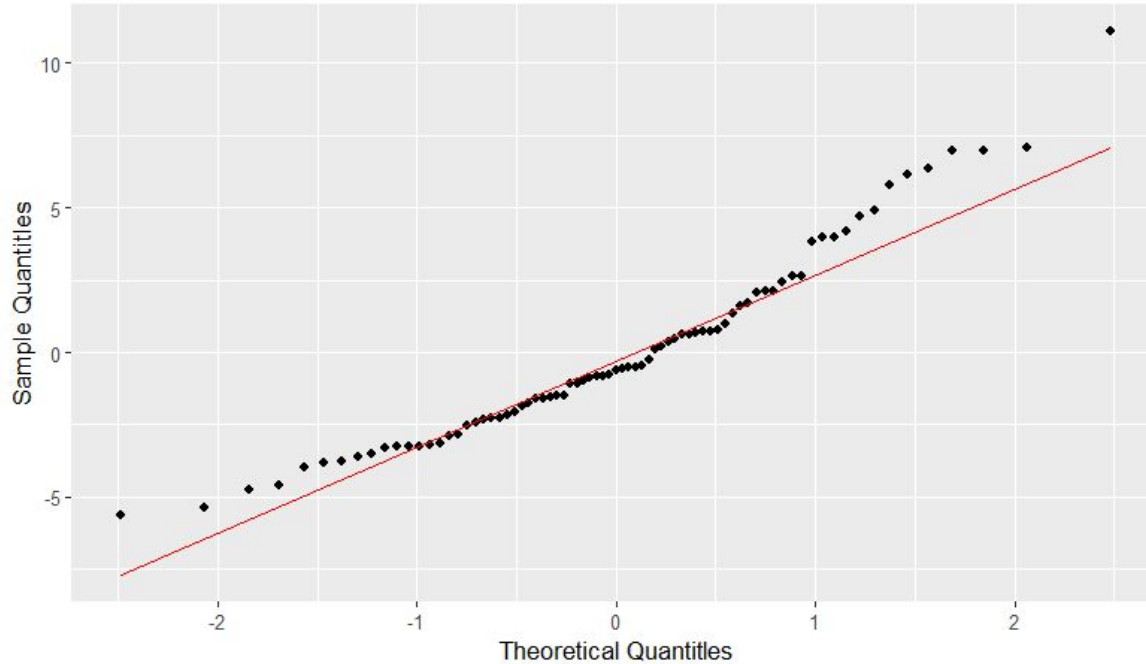
-Data points are around the red line.

-Points considerably spread.



Step 3

Q-Q plot showing Residuals of model 2



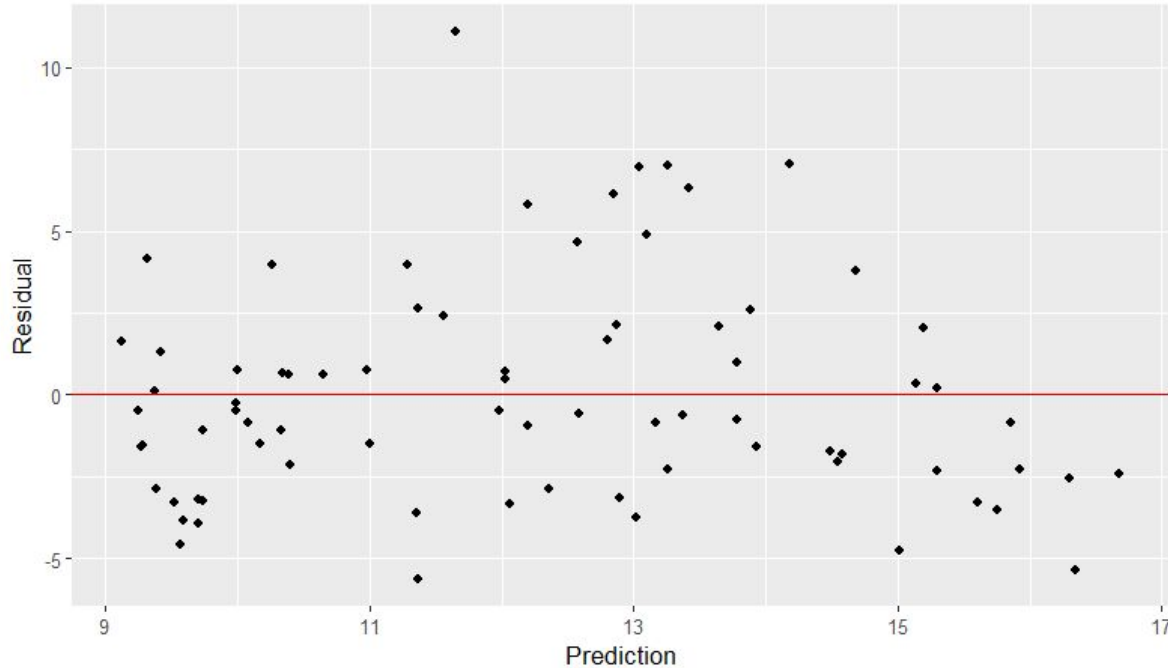
-Most points align with the theoretical normal line

-Outliers at the each end of the plot.



Step 3

Residuals vs predicted plot of model 2

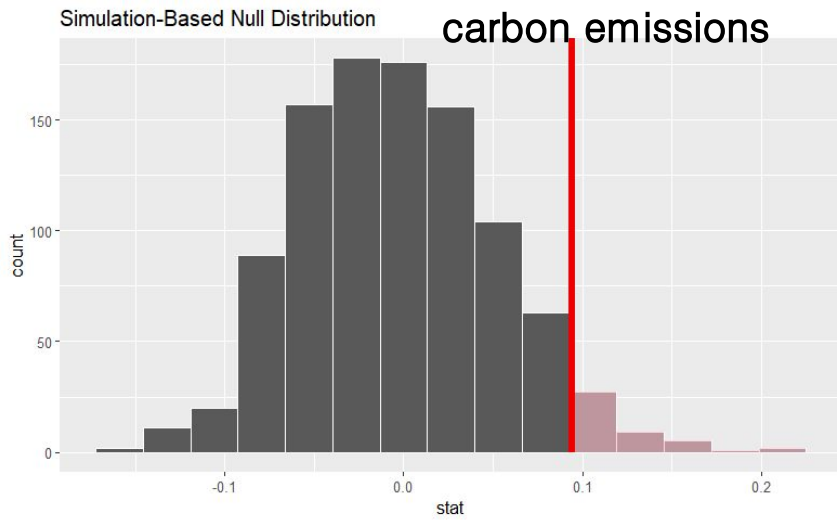


- The reference line will always be at 0
- The residuals are dispersed around the zero line
- Points considerably spread.



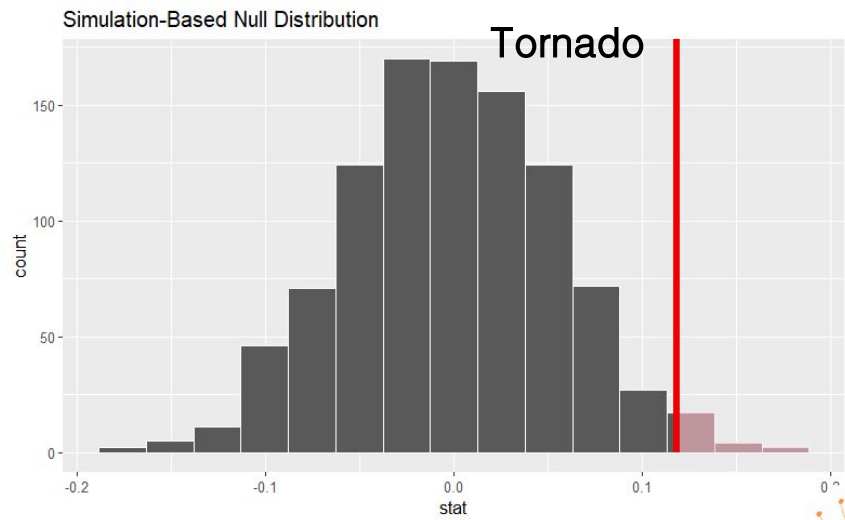
Conclusion





→ p-value for carbon emissions: 0.043

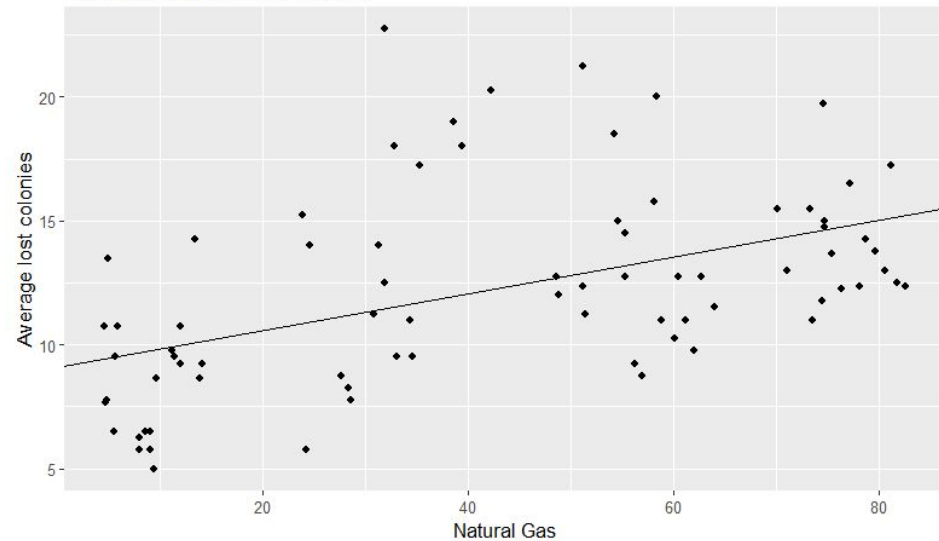
Hypothesis Test Result



→ p-value for Tornado: 0.015

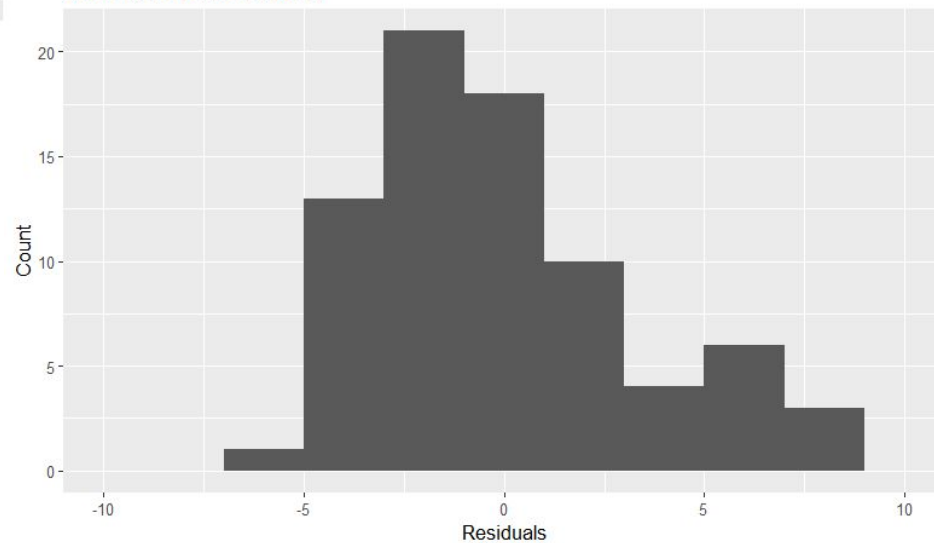


Linearity condition of model 1

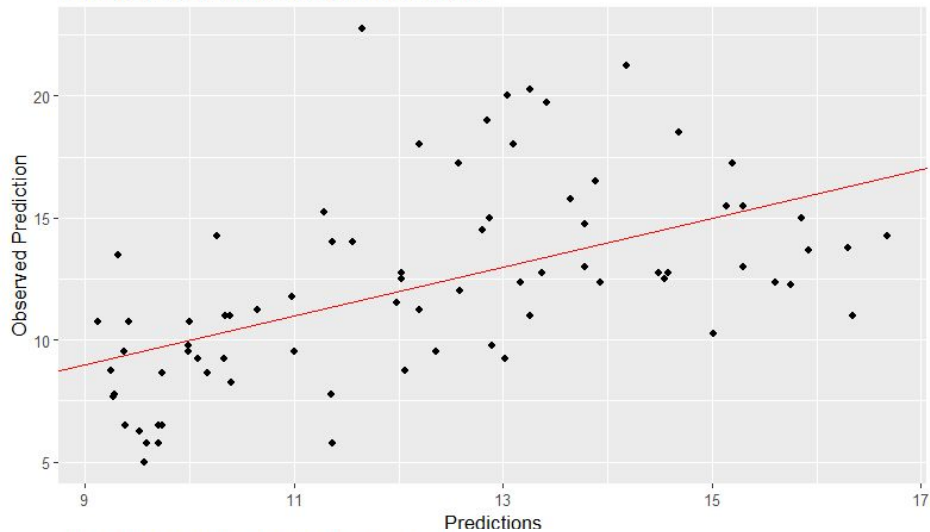


Modeling visualization

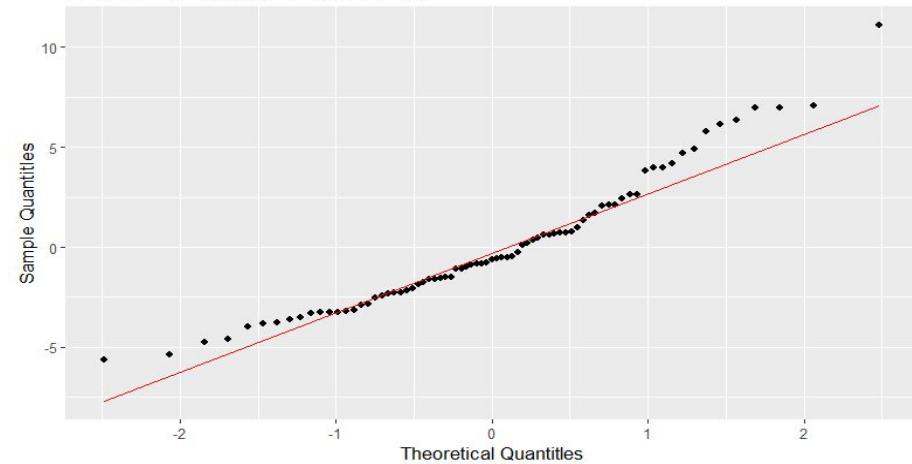
Residual Count of model



Observed Prediction of factors of model 2

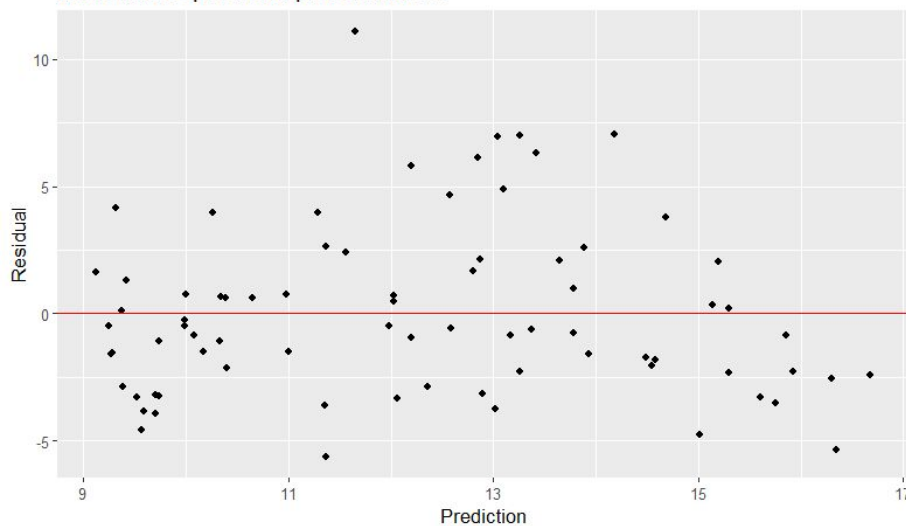


Q-Q plot showing Residuals of model 2



Multiple Linear Regression

Residuals vs predicted plot of model 2





<Key insights>

Carbon emissions

- The significant p-value of 0.043, coupled with the scatter plot and residual analysis indicating a weak but present linear relationship, suggests that increased carbon emissions contribute to the reduction in honey bee populations.

Tornado frequency

- The significant p-value of 0.015, along with the observed deviations in the scatter plot and residual patterns, indicates that changes in tornado frequency also impact honey bee populations.





Implication

Why they are important

Environmental Policy

- Tornado frequency & carbon emission have impact on bee population, so reducing carbon emissions is crucial to protect overall ecological health which might indirectly affect bees.

Research Direction

- Given the impact of external factors like carbon emissions and tornado frequency, we should continue Investigating other potential factors (internal) that might affect honey bee populations
- **ex) Diseases** like Varroa mites, Nosema / **Pesticide** use / **Habitat loss**



Implication

What they can be used for

Conservation Strategies

(Focus on main factors causing honey bee population decline)

- Pesticide management
- Habitat restoration
- Carbon Emission Reduction



Thank You

